Enumeration

Combinatorial Algorithms

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Enumeration



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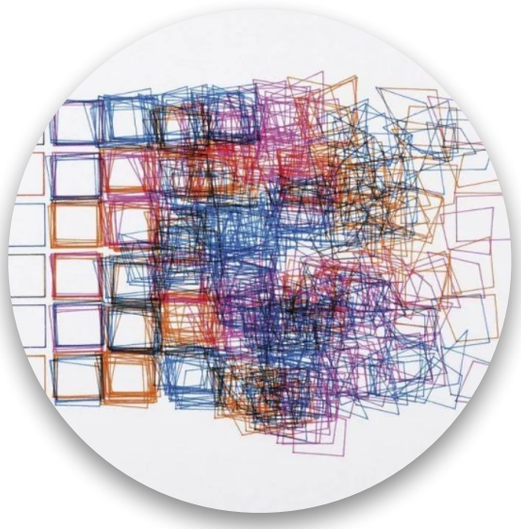
Let us begin by looking into the enumeration of:

- subsets

As discussed, enumeration comes with rank, unrank, and successor functions, associated to particular orderings.

In this class, we will cover the enumeration of sets in lexicographic order, that is, the order in which subsets would appear if we looked them up in a "dictionary" of their characteristic vectors.

Enumeration



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Our collection of structures is

$$S = 2^{[0:n)}$$

the collection of all subsets of $[0 : n)$ which has $\#S = 2^n$ elements.

In the lexicographic order on S, the general framework specializes as:

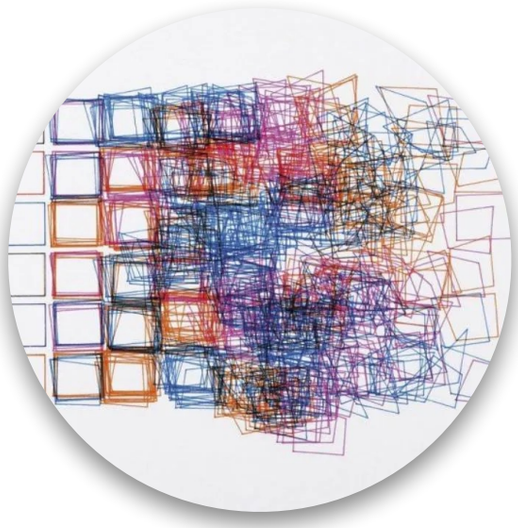
(a) rank takes a subset $A \in 2^{[0:n)}$ and returns its position in $[0 : 2^n)$

(b) unrank takes a position $r \in [0 : n)$ and returns the subset $A \in 2^{[0:n)}$ at that position

(c) successor takes a subset $A \in 2^{[0:n)}$ and returns the next subset $B \in 2^{[0:n)}$ in lexicographic order

To make these concrete, we first need a way to represent subsets as sequences.

Subsets



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We describing how to generate the members of $S = 2^{[0:n)}$ in lexicographical order.

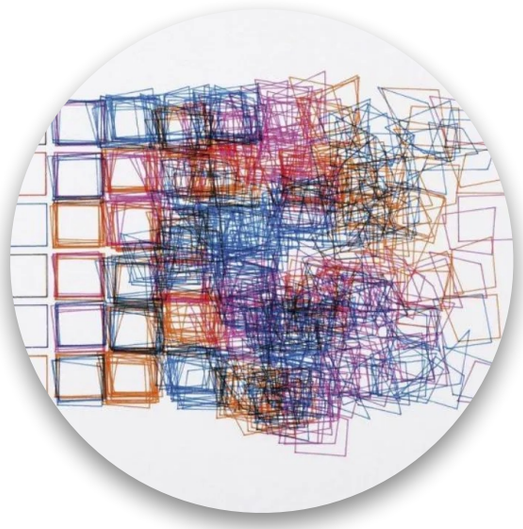
The characteristic vector of a set $A \in S$ is a sequence $\chi(A) \in \{0,1\}^n$ such that for all indices $i \in [0 : n)$ we have:

$$\chi(A)[i] = \begin{cases} 1 & \text{if } (n-(i+1)) \in A \\ 0 & \text{otherwise} \end{cases}$$

The sequence representation of the characteristic function of a set is convenient for the purposes of the algorithms we describe next.

The lexicographic order on S is the lexicographic order of the characteristic vectors. If we think of characteristic vectors as the binary representations of the numbers in the set $[0 : 2^n)$, then, the lexicographic order is the usual total order on natural numbers.

Example



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Let $n = 3$. Then, $S = 2^{[0:3)} = 2^{\{0,1,2\}}$ with $\#S = 2^3 = 8$ elements.

The lexicographic order on the characteristic vectors of S is:

rank	subset	χ	as a number
0	$\varnothing$	`000	0
1	$\{0\}$	`001	1
2	$\{1\}$	`010	2
3	$\{0,1\}$	`011	3
4	$\{2\}$	`100	4
5	$\{0,2\}$	`101	5
6	$\{1,2\}$	`110	6
7	$\{1,2,3\}$	`111	7

In this table: $\chi(\{0,2\})[0] = 1$ because $n - (0 + 1) = 2 \in A$. In turn, $\chi(\{0,2\})[1] = 0$ because $n - (1 + 1) = 1 \notin A$. Finally, we have $\chi(\{0,2\})[2] = 1$ because $n - (2 + 1) = 0 \in A$.

Then, we have $\mathbf{rank}(\{0,2\}) = \mathbf{int}(101) = 5$, where $\mathbf{int}$ is the value of a binary number as a positive integer.

Subsets



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The ranking and unranking algorithms for lexicographic generation of subsets are very simple.

First, ranking a subset $A \in 2^{[0:n)}$ consists of computing the natural number whose binary representation is $\chi(A)$, that is:

$$\text{rank}(A) = \text{int}(\chi(A))$$

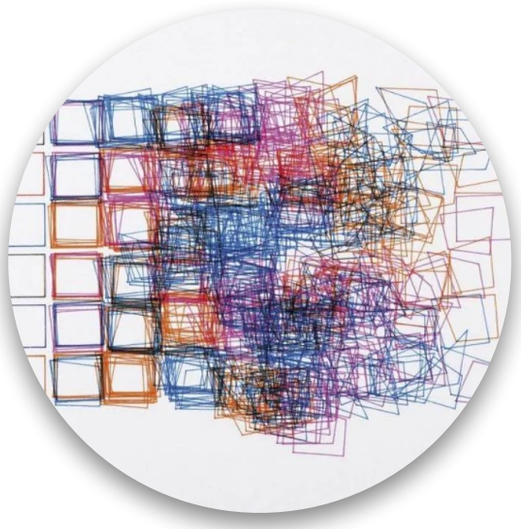
In turn, unranking a position $r \in [0 : 2^n)$ consists of recovering the subset $A \in 2^{[0:n)}$ whose characteristic vector is the n-bit binary representation of r:

$$\text{unrank}(r) = \{n - (i + 1) \,|\, \chi[i] = 1\}$$

where $\chi = \text{bin}(r)$ and $\text{bin}(i)$ is the n-bit binary representation of i.

In the example with $n = 3$: $\text{bin}(5) = 101$, and $\text{unrank}(5) = \{0,2\}$.

Subsets



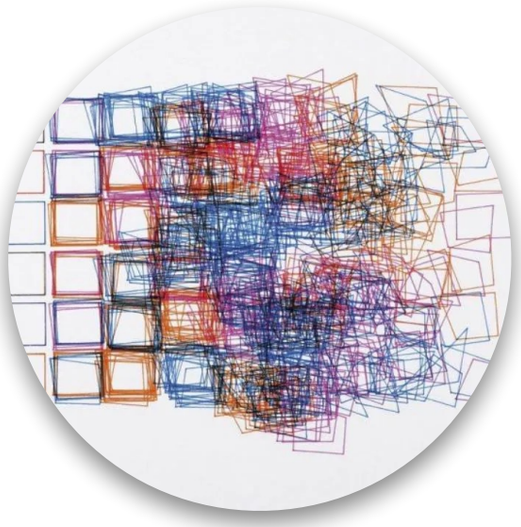
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```
def rank(A: set, n: int) -> int:
    """
    an implementation of the rank algorithm
    """
    # require: A in 2^[0:n)


    ...


    return r
    # ensure: r = int(X(A))
```

Subsets



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```
def unrank(r: int, n: int) -> set:
    """
    an implementation of the unrank algorithm
    """
    # require: r in [0:2^n]


    ...


    return A
    # ensure: int(X(A)) = r
```

Subsets



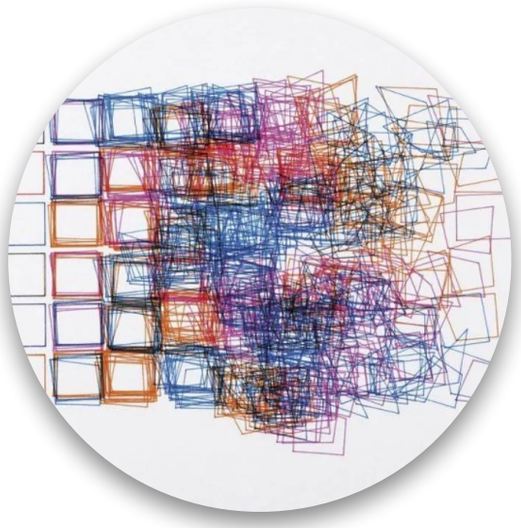
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```
def successor(A: set, n: int) -> set:
    """
    an implementation of the successor algorithm
    """
    # require: A: 2^[0:n)


    ...


    return B
    # ensure: X(B) = X(A)+1
```

Summary



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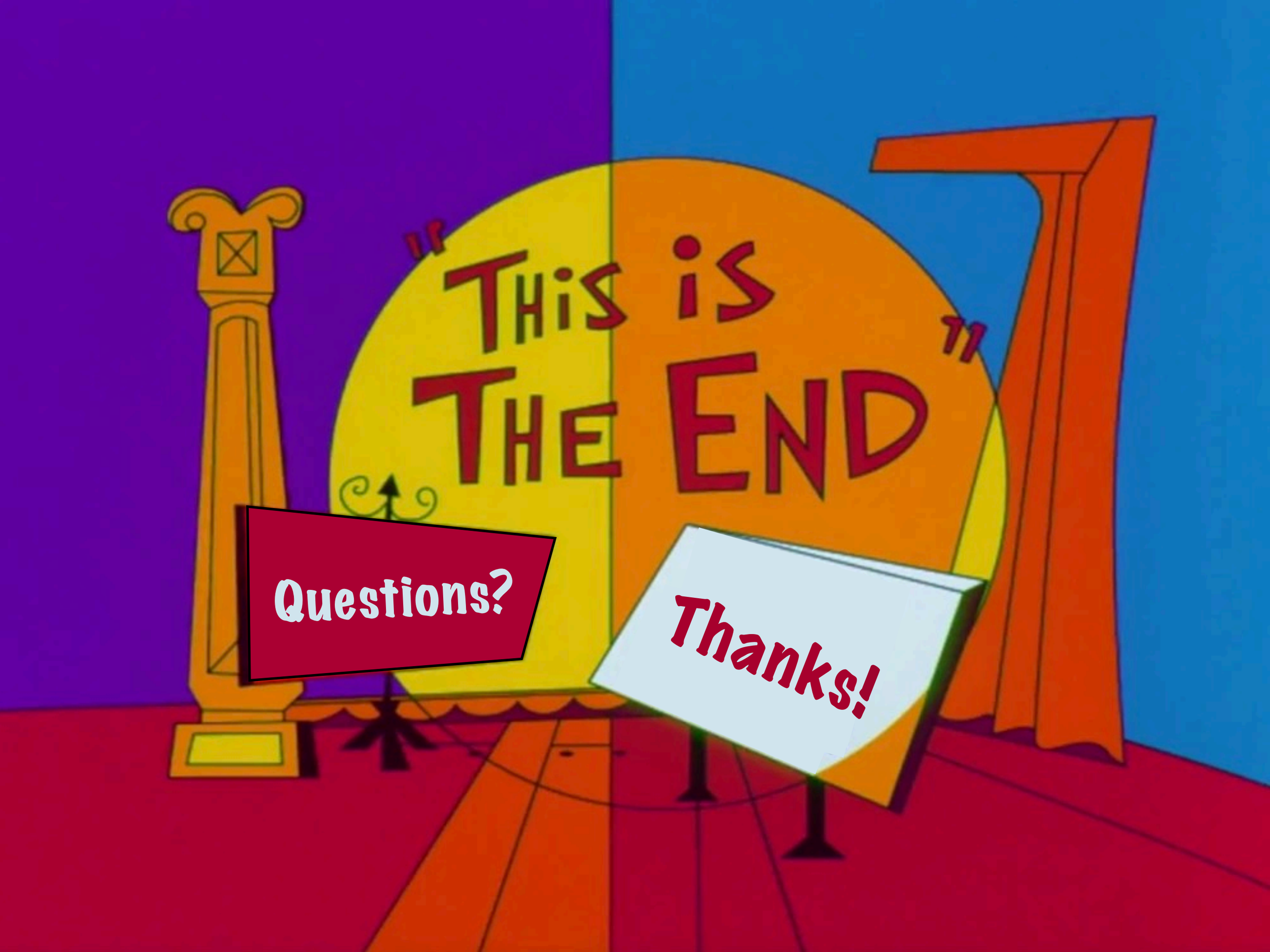
The set of structures is $S = 2^{[0:n)}$, with 2^n elements ordered.

The characteristic vector $\chi(A) \in \{0,1\}^n$ represents a subset A as an n-bit string, with the most significant bit corresponding to the largest element in the set.

The lexicographic order on S is the lexicographic order on the characteristic vectors represented as n-bit strings, or analogously the natural order on the binary representation if the numbers in the set $[0 : 2^n - 1)$.

Ranking reads a set as a characteristic vector, and returns a natural number. Unranking reads a natural number as its binary representation, that is, a characteristic vector, and returns its corresponding set.

Successor increments the characteristic vector by 1: the next subset in lexicographic order is the one whose rank is one higher, circling back to the first element.


"This is The End"
Questions?
Thanks!